

Lab 1 – Intrinsic Explainable Models

Group 1 ($S = 1$)

Luís Forti - 3981932
Marcell Slyuch - 3992616

Contents

Q1) Load the data	2
Q2) Perform basic exploratory data analysis	3
Q3) Train intrinsic explainable models	6
Q4) Explaining the logistic model	7
Q5) Explaining the decision tree	11
Q6) Explaining kNN	14
Q7) Reflection	16

Q1) Load the data

a) [Basic]

Load the data file of your group (`personal_data_path`). What is the file format? Which method of `pandas` reads this format? Which values of `sep`, `header` and `decimal` does your file need, and why?

Notebook output – path and first lines of the raw file:

```
data\heart_group_01.csv
age,sex,cp,trestbps,chol,fbs,restecg,thalach,exang,oldpeak,slope,ca,thal,target
64,0,0,180,325,0,1,154,1,0.0,2,0,2,1
57,1,0,132,207,0,1,168,1,0.0,2,0,3,1
```

`df.info()` with `pd.read_csv(path)`: 273 entries, 14 columns, no missing values.

Column	Non-Null	Dtype
age	273	int64
sex	273	int64
cp	273	int64
trestbps	273	int64
chol	273	int64
fbs	273	int64
restecg	273	int64
thalach	273	int64
exang	273	int64
oldpeak	273	float64
slope	273	int64
ca	273	int64
thal	273	int64
target	273	int64

Answer:

The file format is `.csv`, as can be seen in the print `"data\heart_group_01.csv"`. `Pandas` reads it through the method `read_csv(file_path)`. It doesn't require to explicitly define `sep`, `header` or `decimal`. Both `sep` and `decimal` standard attributions (`,` and `.`, respectively) will suffice the `.csv`. The `header` also doesn't require attribution, since the first line of the document has its definition. `read_csv` will automatically read it and assign to the table.

b) [Basic]

For each of the 13 attributes, say if it is numerical, binary or categorical. Give your answer as a table.

Answer:

Attribute	Type
age	Numerical
sex	Binary
cp	Categorical
trestbps	Numerical
chol	Numerical
fbs	Binary
restecg	Categorical
thalach	Numerical
exang	Binary
oldpeak	Numerical
slope	Categorical
ca	Categorical
thal	Categorical

c) [Regular]

Load the file once more with a wrong value for `sep` or `decimal`. What happens to the columns? How can you see the problem with `df.info()`?

Notebook output – `df.info()` and `df.head()` with `sep=".", decimal=","`:

```
<class 'pandas.core.frame.DataFrame'>
Index: 273 entries, 64,0,0,180,325,0,1,154,1,0 to 68,1,2,180,274,1,0,150,1,1
Data columns (total 1 columns):
 # Column Non-Null Count Dtype
-----
0 age,sex,cp,trestbps,chol,fbs,restecg,thalach,exang,oldpeak,slope,ca,thal,target 273 non-null object
dtypes: object(1)

      age,sex,cp,trestbps,chol,fbs,restecg,thalach,exang,oldpeak,slope,ca,thal,target
64,0,0,180,325,0,1,154,1,0 0,2,0,2,1
57,1,0,132,207,0,1,168,1,0 0,2,0,3,1
44,1,2,140,235,0,0,180,0,0 0,2,0,2,1
69,1,2,140,254,0,0,146,0,2 0,1,3,3,0
58,0,2,120,340,0,1,172,0,0 0,2,0,2,1
```

Answer:

When swapping the decimal and the separator the code still ran, but the resulting table was completely wrong. All of the header is now a single column and the data is destroyed. The rows' names are a sequence of numbers separated by commas and so is their values. `df.info()` can demonstrate so by looking at the columns' names and the rows' indexes.

Q2) Perform basic exploratory data analysis

a) [Basic]

Before you compute anything, write down the three attributes you expect to be most related to heart disease. Then: if you were limited to 5 of the 13 attributes, how would you choose them without a trained model? Show the result as a heatmap of the correlation matrix.

Notebook output:

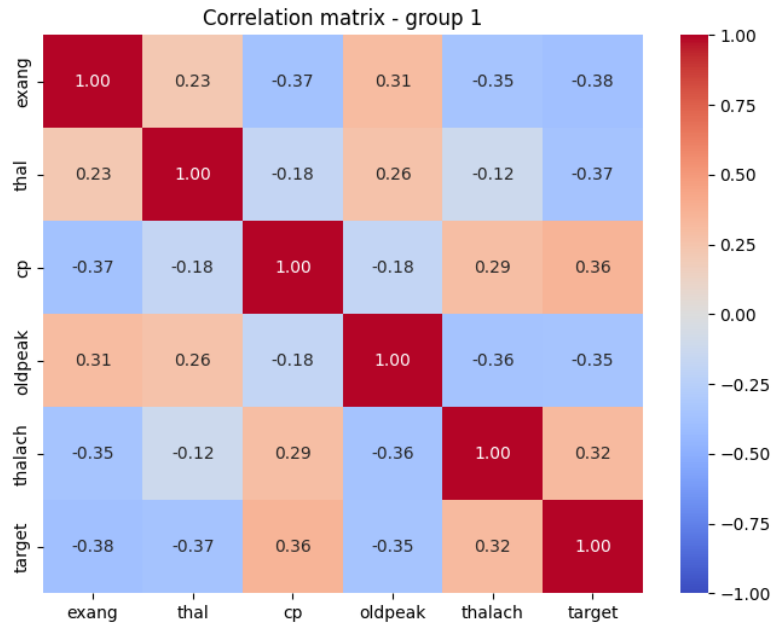


Figure 1: Correlation matrix of the 5 attributes most correlated with the target – group 1

Answer:

[age, restecg, oldpeak]. To choose my top 5, I would create a correlation matrix of all the columns with "target", as shown above.

b) [Basic]

Which 5 attributes have the strongest correlation with the target in your data? Give the values. Was your expectation right?

Notebook output – correlation of every attribute with `target` (sorted by $|r|$):

Rank	Attribute	Correlation
1	exang	-0.384595
2	thal	-0.365281
3	cp	0.358467
4	oldpeak	-0.353332
5	thalach	0.318367
6	slope	0.304794
7	sex	-0.290582
8	ca	-0.272019
9	trestbps	-0.100684
10	fbs	-0.084737
11	restecg	0.068386
12	chol	-0.044971
13	age	-0.020541

Answer:

The only attribute that I expected here was "oldpeak", missing all of the others.

Attribute	Correlation
exang	-0.384595
thal	-0.365281
cp	0.358467
oldpeak	-0.353332
thalach	0.318367

c) [Basic]

How does this help to explain the model later? What does correlation *not* tell us?

Answer:

It creates a baseline of expectations: the higher the correlation, the more likely an attribute is to have an impact on the target. However, correlation has several limitations. It doesn't prove that an attribute causes heart disease, doesn't capture complex or non-linear relationships, and doesn't account for interactions between multiple attributes. It also doesn't show how important an attribute will be to the final trained model or how accurately the model will predict heart disease.

d) [Regular]

Correlation is not a good measure for categorical attributes such as `cp` and `thal`. Explain why. Compute the mutual information of each attribute with the target and compare the ranking with the correlation ranking.

Notebook output – correlation ranking vs. mutual information ranking:

Rank	Correlation	r	Mutual information	MI
1	exang	-0.384595	thal	0.155551
2	thal	-0.365281	cp	0.097527
3	cp	0.358467	ca	0.076807
4	oldpeak	-0.353332	oldpeak	0.075102
5	thalach	0.318367	exang	0.073049
6	slope	0.304794	slope	0.064149
7	sex	-0.290582	sex	0.045476
8	ca	-0.272019	chol	0.040756
9	trestbps	-0.100684	thalach	0.018014
10	fbs	-0.084737	restecg	0.010406
11	restecg	0.068386	fbs	0.003504
12	chol	-0.044971	age	0.000000
13	age	-0.020541	trestbps	0.000000

Answer:

Correlation is not ideal for categorical attributes such as `cp` and `thal` because their numerical values are codes for categories, not actual numerical quantities. For example, `cp = 3` is not necessarily "more" chest pain than `cp = 1`. Pearson correlation assumes that the numerical differences between values have meaning.

The rankings differ considerably between correlation and mutual information. Correlation ranks `exang`, `thal`, and `cp` highest, while mutual information ranks `thal`, `cp` and `ca` highest. This shows that the choice of relationship measure can affect which attributes appear most relevant. Mutual information is particularly useful for categorical attributes because it does not assume that their numerical codes represent meaningful distances or that their relationship with the target is linear.

e) [Advanced]

Find the two attributes that are most correlated with each other (not with the target). Train a logistic model with both attributes, and then with only one of them. How do the coefficients change? Explain why.

Notebook output – highest correlated pair: (oldpeak, slope), $r = -0.5707412531632219$.

Logistic regression	β_{oldpeak}	β_{slope}
Both attributes	-0.51599604	0.56414674
Only oldpeak	-0.66822531	–
Only slope	–	1.04894894

Answer:

"oldpeak" and "slope" have a correlation of approximately -0.571. When used individually, their coefficients are -0.668 and 1.049, respectively. When both are included, the coefficients decrease to -0.516 and 0.564. This happens because the attributes contain overlapping information. When they are used together, the model can account for the information shared between them, reducing the individual contribution assigned to each coefficient. This is an example of multicollinearity.

Q3) Train intrinsic explainable models

a) [Basic]

Complete the preprocessing below (which attributes are scaled, one-hot encoded or kept as they are). Train a logistic model, a decision tree and a kNN model. Report the accuracy and ROC-AUC of each model on your test set.

Notebook output – preprocessing and test-set metrics:

List	Attributes	Transformation
num_cols	age, trestbps, chol, thalach, oldpeak	StandardScaler
cat_cols	cp, restecg, slope, ca, thal	OneHotEncoder(drop="first")
bin_cols	sex, fbs, exang	passthrough

```
logistic:
  Accuracy: 0.8116
  ROC-AUC: 0.8882
tree:
  Accuracy: 0.6957
  ROC-AUC: 0.6577
kNN:
  Accuracy: 0.7681
  ROC-AUC: 0.8577
```

Your patient is row 1 of the test set:

Table 1: The patient (row $p = 1$ of the test set) – group 1

idx	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal
201	65	1	3	138	282	1	0	174	0	1.4	1	1	2

Answer:

Model	Accuracy	ROC-AUC
Logistic Regression	0.8116	0.8882
Decision Tree	0.6957	0.6577
kNN	0.7681	0.8577

b) [Basic]

Why do we scale the attributes for the logistic model and kNN, but not for the decision tree?

Answer:

Logistic regression and kNN benefit from scaling because their calculations are affected by the magnitude of the attributes. kNN uses distances directly, while scaling also helps logistic regression optimization. Decision trees do not need scaling because they split attributes based on thresholds and are therefore not affected by differences in feature scales.

Q4) Explaining the logistic model

a) [Basic]

List the parameters that were fit in the logistic model and explain what they represent.

Notebook output – intercept: 1.6306198087756152; the coefficients are listed in b).

Answer:

Parameter	Coefficient	Interpretation
Intercept	1.631	Log-odds of heart disease when all numerical attributes are at their mean (scaled value 0), binary attributes are 0 and categorical attributes are at their reference category.
age	0.749	Change in log-odds for a 1 SD increase in age.
trestbps	-0.204	Change in log-odds for a 1 SD increase in resting blood pressure.
chol	-0.120	Change in log-odds for a 1 SD increase in cholesterol.
thalach	0.231	Change in log-odds for a 1 SD increase in maximum heart rate.
oldpeak	-0.242	Change in log-odds for a 1 SD increase in ST depression.
sex	-0.910	Change in log-odds for sex = 1 compared with sex = 0.
fbs	0.033	Change in log-odds for fbs = 1 compared with fbs = 0.
exang	-0.659	Change in log-odds for exang = 1 compared with exang = 0.
cp_1.0 / cp_2.0 / cp_3.0	-0.076 / 0.723 / 0.853	Change in log-odds for chest pain type 1 / 2 / 3 compared with type 0.
restecg_1.0 / restecg_2.0	0.502 / -0.658	Change in log-odds for restecg 1 / 2 compared with restecg 0.
slope_1.0 / slope_2.0	-0.493 / 0.391	Change in log-odds for slope 1 / 2 compared with slope 0.
ca_1.0 / ca_2.0 / ca_3.0 / ca_4.0	-1.967 / -1.099 / 0.623 / 0.228	Change in log-odds for ca = 1 / 2 / 3 / 4 compared with ca = 0.
thal_1.0 / thal_2.0 / thal_3.0	0.061 / 1.008 / -0.607	Change in log-odds for thal = 1 / 2 / 3 compared with thal = 0.

b) [Basic]

Compute the odds ratio e^β of each attribute. For the three most important attributes, write one sentence each, using your own numbers. Example: "if `thalach` increases by one standard deviation, the odds of heart disease are multiplied by ...".

Notebook output – coefficients and odds ratios e^β , sorted by $|\beta|$:

Attribute	β	e^β	Attribute	β	e^β
ca_1.0	−1.967420	0.139817	restecg_1.0	0.502469	1.652797
ca_2.0	−1.099469	0.333048	slope_1.0	−0.492923	0.610838
thal_2.0	1.008230	2.740746	slope_2.0	0.391033	1.478508
sex	−0.909961	0.402540	oldpeak	−0.241670	0.785315
cp_3.0	0.852564	2.345653	thalach	0.231488	1.260474
age	0.748664	2.114174	ca_4.0	0.228161	1.256288
cp_2.0	0.723376	2.061380	trestbps	−0.203674	0.815729
exang	−0.658642	0.517554	chol	−0.119996	0.886924
restecg_2.0	−0.657518	0.518136	cp_1.0	−0.076016	0.926801
ca_3.0	−0.623320	0.536162	thal_1.0	0.060988	1.062886
thal_3.0	−0.606828	0.545077	fbs	0.032866	1.033412

Answer:

- `ca_1.0`: For patients in the `ca=1` category, the odds of heart disease are multiplied by 0.140 compared with the reference `ca` category, holding other variables constant.
- `ca_2.0`: For patients in the `ca=2` category, the odds of heart disease are multiplied by 0.333 compared with the reference `ca` category, holding other variables constant.
- `thal_2.0`: For patients in the `thal=2` category, the odds of heart disease are multiplied by 2.741 compared with the reference `thal` category, holding other variables constant.

c) [Basic]

Compute the predicted probability for your patient by hand, using the intercept, the coefficients and the scaled values of your patient. Show the calculation and compare it with `predict_proba`.

Notebook output – scaled values of the patient and their contributions $\beta_j x_j$ (attributes with $x_j = 0$ contribute nothing and are omitted):

Attribute	β_j	x_j (scaled)	$\beta_j x_j$
age	0.748664	1.16677	0.873519
trestbps	−0.203674	0.358054	−0.072926
chol	−0.119996	0.755953	−0.090711
thalach	0.231488	1.05406	0.244002
oldpeak	−0.241670	0.272338	−0.065816
cp_3.0	0.852564	1	0.852564
slope_1.0	−0.492923	1	−0.492923
ca_1.0	−1.967420	1	−1.967420
thal_2.0	1.008230	1	1.008230
sex	−0.909961	1	−0.909961
fbs	0.032866	1	0.032866

$$\begin{aligned}\text{Intercept} &= 1.630620 \\ z &= \beta_0 + \sum_j \beta_j x_j = 1.042043 \\ p &= \frac{1}{1 + e^{-z}} = 0.739244 \quad (\text{by hand}) \\ p &= 0.739244 \quad (\text{predict_proba})\end{aligned}$$

Answer:

The predicted probability calculated by hand is 0.739244, which is identical to the probability obtained using predict_proba. Therefore, the manual calculation correctly reproduces the logistic regression model's prediction for the patient. The model estimates a 73.92% probability of heart disease for this patient.

d) [Advanced]

Train the model once on unscaled and once on scaled data. Compare the order of the attributes by $|\beta|$. Why can we not compare coefficients of unscaled data?

Notebook output – attributes ordered by $|\beta|$ (numerical attributes in bold):

Rank	Scaled data	β	Unscaled data	β
1	ca_1.0	-1.967420	ca_1.0	-1.993166
2	ca_2.0	-1.099469	ca_2.0	-1.132378
3	thal_2.0	1.008230	thal_2.0	1.015537
4	sex	-0.909961	sex	-0.919523
5	cp_3.0	0.852564	cp_3.0	0.849967
6	age	0.748664	cp_2.0	0.712870
7	cp_2.0	0.723376	exang	-0.646508
8	exang	-0.658642	restecg_2.0	-0.645633
9	restecg_2.0	-0.657518	ca_3.0	-0.637270
10	ca_3.0	-0.623320	thal_3.0	-0.605376
11	thal_3.0	-0.606828	restecg_1.0	0.503790
12	restecg_1.0	0.502469	slope_1.0	-0.491168
13	slope_1.0	-0.492923	slope_2.0	0.387265
14	slope_2.0	0.391033	ca_4.0	0.229609
15	oldpeak	-0.241670	oldpeak	-0.199753
16	thalach	0.231488	age	0.088754
17	ca_4.0	0.228161	cp_1.0	-0.080322
18	trestbps	-0.203674	thal_1.0	0.067897
19	chol	-0.119996	fbs	0.037790
20	cp_1.0	-0.076016	trestbps	-0.012483
21	thal_1.0	0.060988	thalach	0.011230
22	fbs	0.032866	chol	-0.002638

Answer:

The ordering of the coefficients changes between the scaled and unscaled models, especially for the numerical attributes. For example, age has $b=0.7487$ when scaled but only $b=0.0888$ when unscaled, while chol changes from $b=-0.1200$ to only $b=-0.0026$. This happens because the unscaled attributes have different units and ranges. A coefficient represents the change in log-odds for a one-unit increase in its corresponding attribute, so the coefficient magnitudes are not directly comparable when the attributes use different scales. After standardization, one unit corresponds to one standard deviation, making the coefficient magnitudes more directly comparable.

e) [Advanced]

Train the model with `cp` as a number (0, 1, 2, 3) and with `cp` one-hot encoded. How does the explanation of `cp` change? Which version makes sense?

Notebook output – `cp` as a number vs. one-hot encoded:

Encoding	Coefficient	β
Numerical (scaled)	<code>cp</code>	0.488972
One-hot	<code>cp_1.0</code>	−0.076016
One-hot	<code>cp_2.0</code>	0.723376
One-hot	<code>cp_3.0</code>	0.852564

Answer:

With `cp` as a number there is only one coefficient, $\beta = 0.489$ per standard deviation. This forces the effect to be linear: going from type 0 to 1 must have the same effect as going from 2 to 3. With one-hot encoding, the model gives `cp_1` = −0.076, `cp_2` = 0.723, `cp_3` = 0.853 compared with `cp_0`. So type 1 behaves almost like type 0, and types 2 and 3 have a much higher risk — not the equal steps that the numerical version assumes. `cp` is a type of chest pain, not a quantity, so the one-hot version makes sense.

f) [Advanced]

Use `statsmodels` to get 95% confidence intervals for the odds ratios. Which intervals contain 1? What does that mean for your explanation?

Notebook output – `statsmodels` logit output:

Attribute	Odds ratio	95% CI lower	95% CI upper	Contains 1
<code>age</code>	3.216077	1.702497	6.075281	no
<code>trestbps</code>	0.693524	0.412684	1.165481	yes
<code>chol</code>	0.832202	0.489626	1.414468	yes
<code>thalach</code>	1.065571	0.560479	2.025843	yes
<code>oldpeak</code>	0.976850	0.540181	1.766511	yes
<code>cp_1.0</code>	0.712184	0.183140	2.769501	yes
<code>cp_2.0</code>	2.435338	0.715026	8.294623	yes
<code>cp_3.0</code>	4.539605	0.556002	37.064665	yes
<code>restecg_1.0</code>	1.812378	0.681550	4.819474	yes
<code>restecg_2.0</code>	0.016420	0.000048	5.651726	yes
<code>slope_1.0</code>	0.774649	0.119890	5.005279	yes
<code>slope_2.0</code>	3.873654	0.428965	34.980016	yes
<code>ca_1.0</code>	0.037973	0.010008	0.144074	no
<code>ca_2.0</code>	0.066817	0.011334	0.393907	no
<code>ca_3.0</code>	0.174849	0.031828	0.960548	no
<code>ca_4.0</code>	2.687318	0.069343	104.144931	yes
<code>thal_1.0</code>	22.440302	0.171581	2934.859546	yes
<code>thal_2.0</code>	36.254982	0.370223	3550.359717	yes
<code>thal_3.0</code>	5.038572	0.052841	480.448750	yes
<code>sex</code>	0.212730	0.057675	0.784638	no
<code>fbs</code>	1.214950	0.362753	4.069171	yes
<code>exang</code>	0.440280	0.154495	1.254712	yes

Answer:

The 95% confidence intervals that don't contain 1 are those of age, ca_1.0, ca_2.0, ca_3.0 and sex. All other intervals contain 1. For these attributes, we cannot say at the 5 % level whether they increase or decrease the odds, so an explanation based on them (e.g. "thal = 2 multiplies the odds by 2.7") is not statistically supported. thal_2.0 also presented 1 within it's interval, but it goes from 0.37 to 3550. Very wide intervals (thal, restecg_2.0, ca_4.0) come from categories with very few patients. The statsmodels model is also not regularised, which is why its odds ratios differ from the scikit-learn ones. Containing 1 does not prove that the attribute has no effect, only that the data is not enough to show it.

Q5) Explaining the decision tree

a) [Basic]

Plot the trained tree so that a person can use it to explain a decision. Follow the path of your patient through the tree and write it as one if-then rule.

Notebook output:

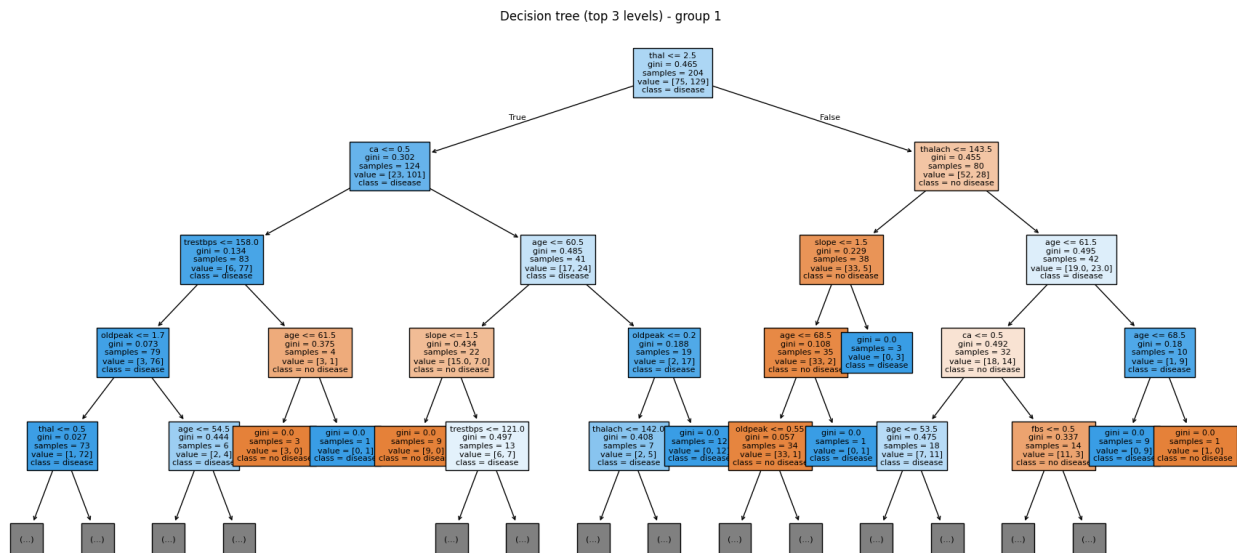


Figure 2: Decision tree (top levels) – group 1

Nodes visited by the patient (`tree.decision_path(x_p).indices`): [0 1 17 27 33].

Answer:

IF thal <= 2.5 AND ca > 0.5 AND age > 60.5 AND oldpeak > 0.2, THEN class = disease

b) [Regular]

Train trees with the split methods "gini", "entropy" and "log_loss". Plot the accuracies and comment on how the trees differ.

Notebook output:

Criterion	Accuracy	Depth	Leaves	Root split
<code>gini</code>	0.6957	10	34	<code>thal ≤ 2.50</code>
<code>entropy</code>	0.7391	10	33	<code>thal ≤ 2.50</code>
<code>log_loss</code>	0.7391	10	33	<code>thal ≤ 2.50</code>

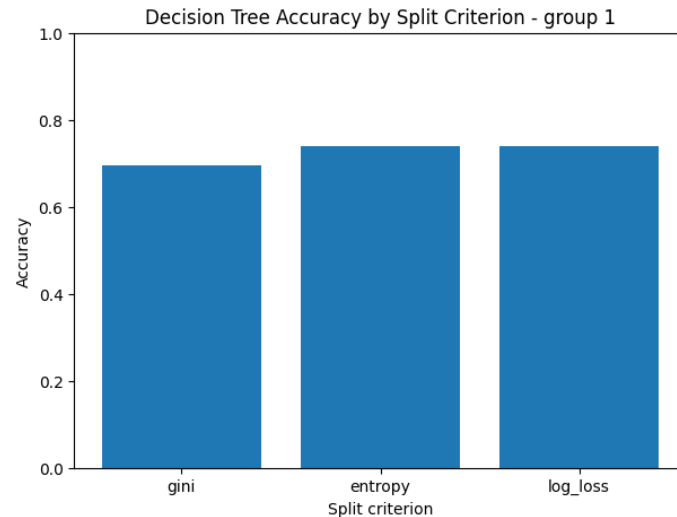


Figure 3: Decision tree accuracy by split criterion – group 1

Answer:

`gini` reaches an accuracy of 0.6957, while `entropy` and `log_loss` both reach 0.7391. In scikit-learn, "`log_loss`" and "`entropy`" are the same criterion (Shannon entropy), so these two trees are identical — same splits, same depth, same leaves. The `gini` tree and the `entropy` tree start with the same root split (`thal ≤ 2.5`) and have the same depth (10), but they differ lower down (34 vs 33 leaves): the two impurity measures rank candidate splits slightly differently, and one different split changes the whole subtree below it. The difference in accuracy on a test set of only 69 patients (3 patients) is small, so it should not be over-interpreted.

c) [Regular]

Take the class counts at the root node of your tree. Compute the Gini index, the entropy and the classification error of the root split by hand. Show the calculation.

Notebook output – calculation by hand. Root counts: [75, 129], total samples: 204.

$$\text{Gini index} = 1 - \left(\frac{75}{204}\right)^2 - \left(\frac{129}{204}\right)^2 = 1 - (0.367647)^2 - (0.632353)^2 = 0.464965$$

$$\text{Entropy} = - (0.367647 \times \log_2(0.367647) + 0.632353 \times \log_2(0.632353)) = 0.948848$$

$$\text{Classification error} = 1 - \max(0.367647, 0.632353) = 0.367647$$

Answer:

The calculation is in the print above.

d) [Regular]

Plot the train and test accuracy for `max_depth` from 1 to 12. Which depth would you show to a doctor? Explain your choice.

Notebook output:

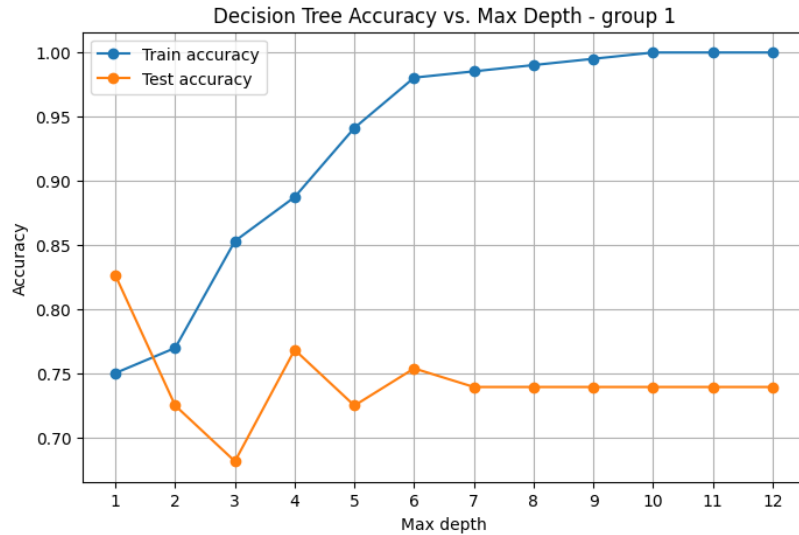


Figure 4: Decision tree train and test accuracy vs. `max_depth` – group 1

Answer:

I would show the tree with a maximum depth of 1 to the doctor. It has the highest test accuracy (82.61%) among the tested depths and is also the simplest and most interpretable tree. As the depth increases, the training accuracy increases considerably, reaching 100%, while the test accuracy does not improve. This suggests that deeper trees are overfitting the training data.

e) [Advanced]

Train a tree with `max_depth=3` on five bootstrap samples of your training data (seeds $S, S+1, \dots, S+4$). Compare the first two levels of the trees. What does this say about the stability of tree explanations?

Notebook output – first two levels of the `max_depth=3` trees (from `export_text`):

Bootstrap	Level 1 (root)	Level 2, left child	Level 2, right child
1	$ca \leq 0.50$	$exang \leq 0.50$	$age \leq 61.50$
2	$exang \leq 0.50$	$ca \leq 0.50$	$age \leq 62.50$
3	$oldpeak \leq 1.90$	$age \leq 60.50$	$age \leq 61.00$
4	$thal \leq 2.50$	$cp \leq 0.50$	$age \leq 61.50$
5	$ca \leq 0.50$	$thal \leq 2.50$	$age \leq 61.50$

Answer:

The first two levels of the trees vary substantially between the five bootstrap samples. The root split changes from `ca` to `exang`, `oldpeak`, or `thal`, while the subsequent splits also differ. This shows that individual decision-tree explanations are relatively unstable: small changes in the training data can produce different decision rules. Therefore, the explanation provided by one particular tree should be interpreted with caution.

Q6) Explaining kNN

a) [Basic]

Plot the accuracy for different values of k (odd values from 1 to 51). Comment on how the explanation changes with different values of k .

Notebook output:

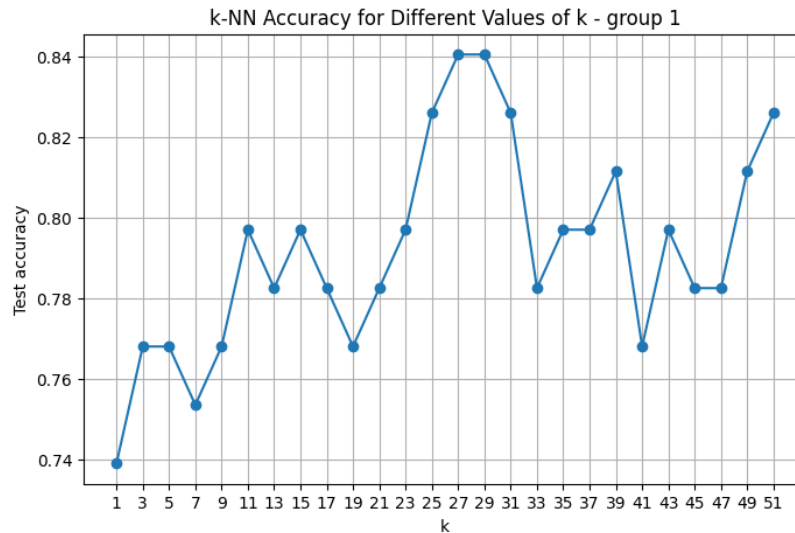


Figure 5: k-NN test accuracy for odd k from 1 to 51 – group 1

Answer:

Small values of k make the prediction depend strongly on a few nearby training observations, making the explanation more local and sensitive to individual observations. As k increases, more observations contribute to the prediction, making the explanation more general and less sensitive to individual observations. In this experiment, the highest test accuracy is 84.06% for $k = 27$ and $k = 29$.

b) [Basic]

Look at the k nearest neighbours of your patient in the training data (code below). Would this convince a doctor? Why or why not?

Notebook output:

Table 3: The patient

idx	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal
201	65	1	3	138	282	1	0	174	0	1.4	1	1	2

Table 4: The $k = 5$ nearest neighbours of the patient (scaled and encoded data)

idx	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target	distance
171	56	1	0	125	249	1	0	144	1	1.2	1	1	2	0	2.356328
69	66	1	0	120	302	0	0	151	0	0.4	1	0	2	1	2.406769
65	63	1	0	130	254	0	0	147	0	1.4	1	1	3	1	2.417764
232	58	1	1	120	284	0	0	160	0	1.8	1	0	2	0	2.460707
44	66	0	2	146	278	0	0	152	0	0.0	1	1	2	1	2.533330

Answer:

Even with scaling and encoding, this would not strongly convince a doctor because the five nearest neighbours are split between 3 patients with target 1 and 2 with target 0, providing no clear consensus. Moreover, Euclidean distance gives equal importance to all features, even though some clinical indicators may be more relevant to heart disease than others. Therefore, k-NN provides a prediction based on similarity to previous patients, but it does not provide an interpretable medical explanation based on pathophysiology or established clinical risk factors.

c) [Regular]

Repeat b) without scaling. Which attribute now decides who the neighbours are? Why?

Notebook output:Table 5: The $k = 5$ nearest neighbours of the patient without scaling

idx	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target	distance
14	67	0	2	152	277	0	1	172	0	0.0	2	1	2	1	15.425952
193	62	0	0	140	268	0	0	160	0	3.6	0	2	2	1	20.392155
47	54	0	1	132	288	1	0	159	1	0.0	2	1	2	1	20.638798
68	51	1	0	140	299	0	1	173	1	1.6	2	0	3	0	22.339203
256	49	0	1	134	271	0	1	162	0	0.0	1	0	2	1	23.344378

Answer:

Without scaling, the neighbours are mainly determined by attributes with larger numerical ranges, particularly chol (cholesterol). This happens because Euclidean distance is calculated directly from the feature values, so a large difference in a high-range attribute contributes much more to the distance than a difference in a low-range or binary attribute. Therefore, without scaling, the nearest neighbours can be determined more by the numerical scale of an attribute than by its actual diagnostic relevance.

d) [Advanced]

Find the smallest change in one attribute of your patient that changes the prediction (a counterfactual). Is this change realistic? You may compare with `dice-ml`.

Notebook output – `dice-ml` counterfactuals varying one numerical attribute at a time, sorted by the standardized change (227 rows, first and last 5 shown):

	attribute	original	counterfactual	absolute_change	standardized_change
100	chol	282.0	255.2	26.8	0.525019
75	chol	282.0	255.1	26.9	0.526978
55	chol	282.0	255.0	27.0	0.528937
91	chol	282.0	254.6	27.4	0.536773
76	chol	282.0	253.6	28.4	0.556363
...	...	...	...	...	...
203	oldpeak	1.4	5.9	4.5	3.720334
218	oldpeak	1.4	6.0	4.6	3.803008
224	oldpeak	1.4	6.1	4.7	3.885682
207	oldpeak	1.4	6.2	4.8	3.968357
210	oldpeak	1.4	6.3	4.9	4.051031

Answer:

The smallest counterfactual found changes chol from 282.0 to 255.2, a decrease of 26.8. After standardization, this corresponds to a change of approximately 0.53 standard deviations. This change is sufficient to change the kNN model's prediction. The change is plausible, as cholesterol levels can change over time, although a decrease of 26.8 is still a substantial change. Therefore, the counterfactual is reasonably realistic as a change that could occur over time, but it should not be interpreted as a direct medical recommendation.

Q7) Reflection

a) [Basic]

Use the decision tree to explain the prediction for your patient (i) to the patient and (ii) to a doctor. What is different between the two explanations?

Answer:

(i) "Based on your assessment, we looked at key indicators of your heart health. Your stress test result (thal = 2.0) showed patterns that require closer attention. Looking further, your test identified the presence of major blood vessels with reduced flow (ca = 1.0), and at 65 years old, your heart's electrical recovery after exertion (oldpeak = 1.4) indicates that your heart muscle is under strain during physical activity. When we combine these factors, the diagnostic model indicates a high likelihood of heart disease."

(ii) "The decision tree classifies row 201 as disease based on the following decision path:

1. thal <= 2.5 (2.0 <= 2.5 -> True)
2. ca <= 0.5 (1.0 > 0.5 -> False)
3. age <= 60.5 (65.0 > 60.5 -> False)
4. oldpeak <= 0.2 (1.4 > 0.2 -> False)

The patient presents with fixed thallium defect criteria, fluoroscopy indicating 1 major vessel colored, age over 60.5, and an ST depression (oldpeak) of 1.4 mm. This combination routes directly to a pure terminal node for coronary artery disease."

The differences between the explanations are in the language, terminology and focus. The patient explanation translates abstract feature names and numerical thresholds (thal, ca, oldpeak, gini) into clinical concepts (blood flow, vessel blockage, heart strain). The doctor explanation uses exact feature variable names, precise numerical cutoffs, and standard medical terminology (e.g., ST depression, fluoroscopy). The patient's communication emphasizes personal health context, while the doctor's communication focuses on diagnostic rigor and decision-path logic for rapid peer verification.

b) [Regular]

Assess the decision tree on completeness, comprehensibility and stability, based on what you saw in this lab.

Answer:

Completeness is high because decision trees exhaustively cover the feature space. Every valid combination of patient inputs follows a deterministic, uninterrupted path through the conditional split nodes until reaching a single leaf prediction, leaving no unhandled cases.

Comprehensibility is high at the top levels but drops to moderate across the full tree. While individual decision pathways are easily translated into intuitive if-then rules, the full-depth tree contains numerous branches and split criteria that become visually cluttered and tedious for a human to trace.

Stability is low due to the inherent sensitivity of unpruned decision trees. Minor variations in the training dataset, such as altering the train-test split or removing a few samples, can radically alter the root-level split selection (e.g., swapping that for another attribute), causing a cascade effect that restructures the entire downstream hierarchy.